

Q2(a)

Newton's Law of Gravitation states that any two particles attract each other with a force that is directly proportional to the product of their masses and inversely proportional to the square of distance between them.

[Do not accept force between two bodies – if two bodies are used, the dimensions must be small relative to their separation]

Q2(b)

The gravitational force which the planet exerts on the satellite provides the centripetal force for circular motion of the satellite.

Hence

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$\text{or } v = \sqrt{\frac{GM}{r}}$$

Q2(c)(i)1

Using $v = \sqrt{\frac{GM}{r}}$, we have

$$\frac{v_A}{v_B} = \sqrt{\frac{r_B}{r_A}} = \sqrt{\frac{2.2 \times 10^{10}}{1.3 \times 10^8}}$$

= 13

Q2(c)(i)2

By Kepler's 3rd Law, $T \propto r^{3/2}$

$$\frac{T_A}{T_B} = \left(\frac{r_A}{r_B} \right)^{\frac{3}{2}} = \left(\frac{1.3 \times 10^8}{2.2 \times 10^{10}} \right)^{\frac{3}{2}}$$

= 4.5 × 10⁻⁴

Q2(c)(ii)

Orbital speed of A is $v_A = r_A \omega = (1.3 \times 10^8)(1.7 \times 10^{-4}) = 2.21 \times 10^4 \text{ m s}^{-1}$

But $\frac{v_A}{v_B} = 13$ or $v_B = \frac{2.21 \times 10^4}{13} = 1.7 \times 10^3 \text{ m s}^{-1}$

